N. Fernando Díaz Morera

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Research Interests

I explore the intersection of abstract mathematical symmetry and digital security. My work applies algebraic groups and combinatorics to design robust, error-correcting codes, while actively engaging undergraduate students in research that translates pure mathematics into practical coding implementations.

Appointments

2025 - Present **Fitchburg State University, Assistant Professor**.

2024 – 2025 **Fitchburg State University, Visiting Assistant Professor**.

Education

2018 – 2024 **Ph.D. Mathematics, Tulane University, USA**.
Thesis title: *Torus actions, spherical varieties, Dyck paths, and shellability*.
Advisor: Prof. Mahir Bilen Can.

2016 – 2018 **M.S. Mathematics, Instituto Politécnico Nacional, México**.
Thesis title: Q_p *spaces on hyperbolic Riemann surfaces*.
Advisor: Prof. Luis M. Tovar Sánchez (R.I.P)

2010 – 2015 **B.S. Mathematics, Universidad Sergio Arboleda, Colombia**.
Thesis title: *Complex structures on compact surfaces*.
Adviser: Prof. Leonardo A. Cano García.

Publications and Preprints

1. B. Adenbaum, N. F. Díaz-Morera, J. Elder, P. E. Harris, M. Lynch, and J. C. M. Mori, *Parking cascades: From the simplest sequence to motzkin and catalan*, 2026. arXiv: 2608.00735 [math.CO]. URL: https://arxiv.org/abs/2608.00735.

2. A. Bingham and N. F. Díaz Morera, "Lexicographic shellability of sects," *Electron. J. Combin.*, vol. 32, no. 2, Paper No. 2.50, 20, 2025, ISSN: 1077-8926. DOI: 10.37236/13631.

3. M. B. Can and N. Diaz Morera, "Nearly toric varieties of type A," *Turkish J. Math.*, vol. 49, no. 1, Art. 6, 65–95, 2025, ISSN: 1300-0098,1303-6149. DOI: 10.55730/1300-0098.3574.

[1] Updated October 27, 2026

Teaching Experience

Fitchburg State University

2026 - Present — **Instructor of Record.** Math-1700 Applied Statistics (Fall), Math-2600 Linear Algebra (Fall), Math-4901 Independent Study (Fall) Math-1240 Calculus I (Spring), Math-1300 Precalculus (Spring).

2025 - 2026 — **Instructor of Record.** Math-1300 Precalculus (Fall and Spring), Math-2600 Linear Algebra (Fall and Spring), Math-1240 Calculus II (Spring), Math-4901 Independent Study (Spring).

2024 - 2025 — **Instructor of Record.** Math-1300 Precalculus (Fall and Spring), Math-1700 Applied Statistics (Fall), Math-2600 Linear Algebra (Spring), Math 3900 Math Seminar (Spring), Math-1800 Business Statistics (Summer).

Tulane University

2023 – 2024 — **Instructor.** Math-1230, Statistics for Scientist (Fall and Spring).

Summer '23 — **Instructor.** Math-1110, Probability & Statistic I.

2022 – 2023 — **T.A.** Math-1221 Calculus II (Spring), Math-1311 Consolidated Calculus I (Fall).

Summer '22 — **Instructor.** Math-1110, Probability & Statistic I.

2021 – 2022 — **Instructor**. Math 1150 Long Calculus I (Spring). **T.A.** Math-3090 Linear Algebra (Fall).

2019 – 2021 — **T.A.** Math-1230, Statistics for Scientist (Fall and Spring).

2018 – 2019 — **T.A**. Math-1210 Calculus I (Fall), Math-2210 Calculus III (Spring).

Conferences

Invited talks

2026 — *Bruhat orders and Shellability*. *M@X — Math @ Xavier Seminar*, Xavier University of Louisiana, New Orleans, USA (April 16).

2025 — *On Shelling a Family of Symmetric Spaces*. AMS Special Session on *Interactions between geometry, combinatorics, and flag varieties*), Saint Louis University, Saint Louis, MO, USA (Oct 18-19).

2023 — *Shellability of symmetric spaces and Bruhat orders*. SIAM Texas-Louisiana Sectional Meeting (SIAM TX-LA), University of Louisiana at Lafayette, LA, USA.

Enumerating spherical Dyck paths and smooth nearly toric varieties. Discrete Mathematics and Computer Sciences (DiscreteMath), Univesidad Nacional, Colombia (virtual).

Spherical partition Schubert varieties and Dyck paths. Southern Regional Algebra Conference (SRAC), Tulane University, New Orleans, USA.

Contributed talks

2026 — *On the pattern* 25314 *in Type* A. Joint Mathematics Meetings (JMM), Washington, D.C., USA (Jan 4–7).

2024 — *The Ubiquity of Pattern Avoidance*. MAA-NORTHEAST: NES/MAA, Bridgewater State University, MA, USA.

Symmetric spaces and shellability. CIMPA: VIII-Encuentro Colombiano de Combinatoria, Universidad del Cauca, Colombia.

2023 — *Nearly toric Schubert varieties and Dyck paths*. 13th SE Lie Theory Workshop-NC State university (Combinatorial Representation Theory of Algebras and Applications), Raleigh, NC, USA.

Dyck paths and nearly toric Schubert varieties. CombinaTexas-TAMU (CombinaTexas), College Station, TX, USA.

Dyck paths and nearly toric Schubert varieties. Joint Mathematics Meetings (JMM), Boston, USA.

Conferences (continued)

2022 *Ding and Schubert Varieties*. Congreso Nacional Sociedad Matemática Mexicana, Gaudalajara-México, (hybrid-video).

2018 *Q_p Spaces on Hyperbolic Riemann Surfaces*. Encuentro Sociedades de Matemáticas de Colombia y México, Barranquilla, Colombia.

Posters

2024 *Lexicographic shellability of sects*. Sagan2024, University of Florida, USA.

Lexicographic shellability of sects. CAAC, LACIM, Montréal, Canada.

2023 *Spherical Dyck paths*. Permutation Pattern 2023, University of Burgundy, Dijon, France.

Nearly toric Schubert varieties and Dyck paths. SLAM, University of North Texas, USA.

2022 *Spherical partition Schubert varieties*. PRIMA-2022, Vancouver, Canada.

Outreach and Service

Administration

2025-Present Faculty **Math Representative** for the **Math Club**, Fitchburg State University.

2024-Present Falcon **Math Seminar** Organizer, Fitchburg State University.

Curriculum Committee Member, Fitchburg State University, Fitchburg, MA, USA.

2026-2026 **Referee** for **Mathematics Magazine Journal**-(MMA) Mathematical Association of Mathematics (Aug. 30).

Volunteer in **Poster Session**-JMM content as a **judge**, Washington D.C., USA (Jan 6).

2025-2025 Volunteer in **First-LEGO** content as a **judge**, Montachusett Regional Vocational Technical High School, MA, USA (Dec 6).

2021-2023 Volunteer in Math For All, a conference on Math Education and Research, Tulane University, New Orleans, USA.

2021-2022 **President** of AMS Student Chapter, Tulane University, LA, USA.

Undergrad Research Trainees

2026- 2027 Vanshika Patel and Aiden Thompson, *Topological Data Analysis Applications*, Fitchburg State University, Fitchburg, MA, USA.

2025-2026 Rojanji Nova and Charles Stevenson, ***Math-4901 (Independent Study)*** *Error Correction Meets Post-Quantum Security: A Study of Lattice-Based Cryptography & Classic McEliece*, Fitchburg State University, Fitchburg, MA, USA (spring '26).

2025- 2026 Rojanji Nova and Charles Stevenson, *Error Correction Meets Post-Quantum Security: A Study of Lattice-Based Cryptography*, Fitchburg State University, Fitchburg, MA, USA.

2020-2021 Alexander Caione, *Introduction to Algebraic Actions*, Tulane University, New Orleans, USA.

Supervision of Undergraduate Students

2025-2026 Rojanji Nova and Charles Stevenson, *When Noise Protects and Destroys: An Undergraduate Exploration*, Fitchburg State University, **Poster Presentation**, Joint Mathematics Meetings (JMM), Washington, D.C., Jan 4-7.

2025-2025 Heiner Jeshua Enciso Gaona, *Una Introducción a Códigos Cuánticos con el Código de Cinco Qubits*, Universidad Sergio Arboleda, **Poster Presentation**, XXIV Congreso Colombiano de Matemáticas, Universidad Tecnológica de Pereira, Pereira, Colombia, Jun 2-6.

Outreach and Service (continued)

2023-2025
- Heiner Jeshua Enciso Gaona, *Teoría de Códigos y Criptografía una introducción con Códigos en Python*, Universidad Sergio Arboleda, Bogotá D.C., Colombia (Coadvised with Dr. Diego Villamizar).
- Henry Mauricio Cañón Cortés, *Quantum Coding Theory*, Universidad Sergio Arboleda, Bogotá D.C., Colombia, *Honors Thesis*, (Coadvised with Dr. Diego Villamizar).

Miscellaneous Experience

Awards & Fellowships

2026
- **Falcon Scholars Undergraduate Research Assistance: Mapping Semantic Space: Using Topological Data Analysis to Expose Failure Hotspots in Vector Embeddings**, Fitchburg State University. Joint research with students *Vanshika Patel* and *Aidan Thompson*.
- **Open Educational Resources Grant** for MATH-1700 course, Fitchburg State University.

2025
- **Project NExT Fellow, Orange '25 Cohort**, Mathematical Association of America (MAA).
- **Travel Grant** for Graduate Workshop on Linear Algebra over Finite Fields & Applications, ICERM, Providence, RI, USA.
- **Open Educational Resources Grant** for MATH-2600 course, Fitchburg State University.
- **Falcon Scholars Undergraduate Research Assistance: Error Correction Meets Post-Quantum Security: A Study of Lattice-Based Cryptography**, Fitchburg State University. Joint research with students *Rohanji Novas* and *Charles Stevenson*.
- **Travel Grant** for Collaborate@ICERM: Structures Of The Weak Order, Providence, RI, USA.

2024
- **Open Educational Resources Grant** for MATH-1300 course, Fitchburg State University.
- **Department Prize for Best Teaching Performance**, Tulane University.
- **Travel Grant** for ECCO, Universidad del Cauca, Cauca, Colombia.

2023
- **Travel Grant** for Schubert Summer School @ UIUC, Champaign, IL, USA.
- **AMS Travel Grant** for Joint Mathematics Meetings (JMM), Boston, MA, USA.

2021
- **Travel Grant** for Mathematical Sciences Research Institute (MSRI): Gauge Theory in Geometry and Topology (Virtual School).
- **Summer Research Support** from the department of Mathematics, Tulane university.

2016
- **Scholarship** from The National Council for Science and Technology (CONACYT). Master's studies at IPN-México (2016-2018).

Skills

Languages — English, Spanish (native).

Coding — Familiar with Python and Sage.

Hobbies — Running, football, cycling, and cooking.